When Should Onion Farmers Sell Their Crop?

A forecast-driven sell-or-store plan for Lasalgaon-belt FPOs

Team Kanda Code · equal contribution, 20% each · <roll>@ds.study.iitm.ac.in · reproducible in Python + Excel

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0.55× → 1.55×	0 t stored	+Rs 4,546/t	1.49 seasons
Apr → Nov price, × the year's mean	best chawl plan (18%/month loss)	cold/IoT gain; won 7 of 8 seasons	payback; ≈Rs 45.5 lakh/yr gross

1 Problem Statement

Onion farmers in Maharashtra sell most of their rabi crop at the April–May harvest low, although November prices beat April's in every year 2014–24, by a median 148%; Lasalgaon farmers have been paid Rs 7–10/kg against a cultivation cost of Rs 22–25/kg [16]. Waiting is not free: stored onions shrink and rot – 54.89% in 120 days in a ventilated *kanda chawl*, only 5.00–7.14% in cold storage [4]. **Decision problem:** a farmer producer organisation (FPO) holds Q = 1,000 t at harvest. How many tonnes should it sell in each month, April to December, so that the price rise beats storage loss and cost – and what is low-loss, sensor-monitored storage worth?

2 Motivation for Problem Selection

Recurring and large. The government bought about 6.4 lakh t of onion for its buffer in 2023–24, yet in 2019 NAFED lost about 53% of a 57,372 t buffer held in traditional storage [16] – consistent with ICAR's chawl loss rate. About 65% of the rabi crop is stored and sold in July–December; 1% more availability goes with a 1.3% fall in onion CPI [10]. **The hardest storage case.** Grains keep and have support prices; tomatoes cannot be stored; onion keeps 4–6 months but rots fast, and its price is highly volatile, most of all at Lasalgaon, Asia's largest onion market [9, 12]. **An Industry 4.0 decision.** The gap between 54.9% and about 7% loss prices controlled, IoT-monitored storage [15] in a tested storage-timing frame [13, 14].

3 Research Questions

RQ1. How strongly do arrivals move the onion price? **RQ2.** Does an optimised storage plan beat selling everything at harvest once losses and costs are counted, and in which months should the FPO sell? **RQ3.** What is low-loss (cold or IoT-monitored) storage worth per tonne, and does it repay its capital? **RQ4 (extension).** Does late-monsoon rain seen by satellite foreshadow the price rise, and what is sensing a store's loss rate worth?

4 Hypotheses, Objectives and Constraints

Hypotheses. *From the approved proposal:* **H1:** price falls significantly as arrivals rise ($\beta < 0$ in a log–log regression); **H2:** an optimised plan with a ventilated chawl earns more than selling everything at harvest; **H3:** low-loss storage raises net revenue per tonne by a material margin over a chawl. *Extension beyond the approved proposal:* **H4:** heavier Sep–Oct rain is followed by a bigger rise from the April price to the Oct–Dec mean.

Objectives. (O1) Build and validate a price benchmark; (O2) test H1; (O3) pick a March-origin forecast on validation seasons only; (O4) solve a linear-programme (LP) sell-or-store plan per scenario with verified shadow prices; (O5) backtest on unseen prices and value low-loss storage; (O6, extension) test H4 and value sensor information. **Constraints.** *In the LP:* stock balance with monthly loss δ; capacity C = 700 t; an April crop-loan sale $\geq L$ = 250 t; a grading-and-trucking cap U = 300 t/month in May–Dec (the April sale goes field-run to traders); all stock sold by December. *On the project:* public secondary data and syllabus methods only; the FPO is a price-taker – 1,000 t is 3.6% of Lasalgaon's mean April arrivals (implied price impact ≈1.59%).

5 Exploratory Data Analysis (EDA)

Data and the Lasalgaon pivot. The main source is an Agmarknet compilation of 965,314 all-India daily mandi records, 2014–2024 [1]. Our planned first-week check found **0 rows** for Lasalgaon APMC, so we built a Maharashtra arrival-weighted modal price, $P_m = \Sigma A_i p_i \,/\, \Sigma A_i$ over market-days i of month m, from 91,898 rows in 96 APMCs (Ahmednagar 38.5%, Pune 18.4%, Solapur 15.2% of weight), dropping 42 rows outside Rs 50–20,000/q and 58 relative outliers (outside 0.2–5× the same-day median). **A data trap:** on 5,672 market-days with several variety rows, 100% repeat the *same* tonnage; summing would double-count 1,979,857 t, so we keep one row per market-day (85,346; 132 months, none missing). **Validation.** Against NHRDF's Lasalgaon series [2] (66 months) r = 0.99 on levels, 0.93 on monthly changes (mean ratio 0.973); against 19 independent 2020–24 Lasalgaon reports (8 from MSAMB [3]) r = 0.90, our level a median 10.2% lower (Lasalgaon is the premium market); Nasik APMC tracks Lasalgaon at r = 0.97. **Weather (extension).** NASA POWER [17] satellite/reanalysis data (≈0.5° grid) give monthly temperature, relative humidity (RH) and bias-corrected rain at Lasalgaon (20.15°N 74.23°E), 1996–2024; RH is 82–85% in Jul–Sep, exactly when onions sit in a chawl – physical support for its high loss rate.

Patterns. Mean Rs 14.88/kg, from Rs 4.51 (Feb 2019) to Rs 50.56 (Dec 2019); coefficient of variation 60.9%, month-on-month s.d. 32.5%; the yearly maximum averages 4.58× the minimum. The seasonal index (price ÷ year mean) bottoms at 0.553 in April and peaks at 1.546 in November (Fig. 1b); November beat April in 11 of 11 years (median rise 148.3%), and the 1996–2013 Lasalgaon index (April 0.609, October 1.491) has the same shape. The 2019, 2020 and 2023 spikes coincide with export bans or duties, imposed when prices surge [8].

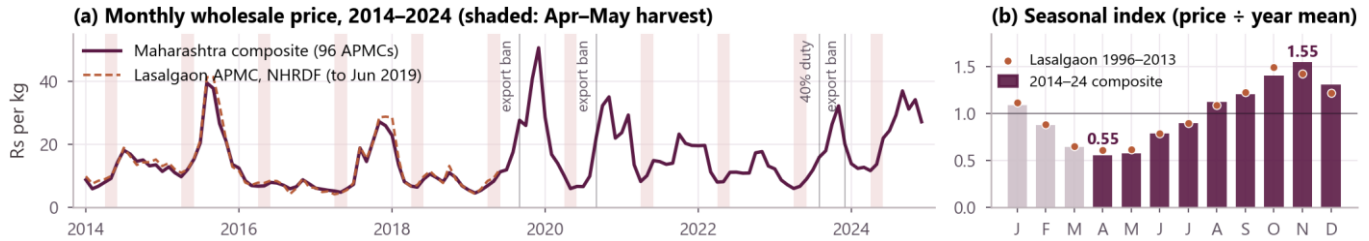


Figure 1. (a) The composite tracks Lasalgaon; April–May is the trough. (b) Seasonal index 2014–24 (bars), Lasalgaon 1996–2013 (dots).

6 Methodology

H1 – regression. OLS of ln P_m on ln A_m with 11 month and 10 year dummies (132 months); A_m is arrivals per market-day, because reporting markets rose from 34.0 to 63.8 a month and raw totals partly measure coverage. Residuals are autocorrelated (Durbin–Watson 0.52), so we use Newey–West (HAC, 3-lag) standard errors, a one-sided test of $\beta < 0$ and robustness checks (normalisations, lags, daily data, reverse regression).

Forecasting protocol. Origin = end of March; forecast April–December (1–9 steps) on an expanding window from 2014. Ten syllabus candidates: naive, seasonal naive, 12-month moving average, seasonal index × moving average or exponential smoothing (index from our series or 28 years with NHRDF 1996–2013), damped Holt–Winters and two regressions. **Selection rule (test held out):** lowest April–December MAPE on validation seasons 2019–22, then one score on test seasons 2023–24. Selected: **regression with trend + month dummies**, $\ln P_t = a + bt + \Sigma\gamma_k D_{k,t}$ (validation MAPE 36.9% vs 42.1% for seasonal naive): beyond accuracy, its month dummies carry the seasonal shape the decision needs, the trend absorbs rising nominal prices, and it is interpretable. The test-best model (28-yr seasonal index × exponential smoothing, 19.6%) was not adopted: rankings are unstable (Holt–Winters 57.4% validation, 23.9% test) and switching after seeing the test would leak it.

Sell-or-store LP. x_t = tonnes sold and s_t = tonnes in store at the end of month t (t = 1 April, …, 9 December). Prices are exogenous (price-taker) and loss is proportional, so the model is a linear programme:

$$\max Z = \Sigma \hat{p}_t x_t - h\Sigma s_t \quad \text{s.t.} \quad s_1 = Q - x_1;\ \ s_t = (1-\delta)s_{t-1} - x_t;\ \ s_t \le C;\ \ x_1 \ge L;\ \ x_t \le U\ (t \ge 2);\ \ s_9 = 0;\ \ x, s \ge 0$$

Here $\delta = 1 - (1 - \lambda_{120})^{1/4}$ converts ICAR's 120-day trial loss λ_{120} to a monthly rate (Table 1). The LP is solved in PuLP (CBC), cross-checked in SciPy/HiGHS (objective gap ≤ Rs 0.01) and Excel Solver; shadow prices are reported only after ±1 t re-solves verify them.

Scenario	Monthly loss δ	Storage cost *h*	Capital cost	Source
A No storage	– (C = 0: sell all in April)	–	–	current practice
B Ventilated kanda chawl	18.05% (54.89% in 120 days)	Rs 250/t/month	Rs 7,000/t	[4], [6], [7]
C Cold / IoT-monitored	1.83% (7.14% in 120 days)	Rs 700/t/month (operating)	Rs 16,667/t	[4], [5], [15]

Table 1. Storage scenarios. All share Q = 1,000 t, C = 700 t, L = 250 t, U = 300 t/month; capital is costed apart from h.

Backtest and evaluation. For each season 2017–24 we forecast with data to March only, solve the LP, then *implement the plan at actual prices*: net = $\Sigma p_t x_t - h\Sigma s_t$, gain = (net − $Q \cdot p_{April}$) / Q per tonne harvested. Benchmarks: rolling monthly re-planning, an ex-ante calendar rule (sell in the top-3 seasonal-index months known at the origin – Sep, Oct, Nov every year) and perfect foresight (the ceiling). One-sided paired t and Wilcoxon tests over 8 seasons; 2023–24 are the clean test seasons (2019–22 also informed model choice). Sensitivity: one parameter at a time plus a δ × h grid. **Industry 4.0 value:** C's gain over B per tonne of capacity against the extra capital, as payback and annuity (recovery factor 0.1627, 10 yr at 10%).

Extension. *H4:* OLS of ln(mean Oct–Dec ÷ April price) on Sep+Oct rain (a crop-calendar window, not searched), one season per observation (n = 29, 1996–2024; NHRDF before 2014), HC1-robust p, leave-one-out and subperiod refits, and a Jun–Sep monsoon placebo. *Value of information:* the store's true δ varies on a grid; a *blind* operator plans with the book δ, a *sensor* one measures δ from May and re-plans monthly, a *calibrated* one knows δ before April – each run with our forecast and with perfect prices (the pure value of loss information). A 1 Nov re-plan on the rain signal (β from earlier seasons only) values a real-time rain feed.

7 Results

Hypothesis	Test	Result	Verdict
H1 arrivals ↓ price	log–log OLS + month, year dummies; HAC; n = 132	β = −0.442, OLS CI [−0.79, −0.10]; HAC one-sided p = 0.007; ≈ 0 with lagged price	**Supported** (association)
H2 chawl storage pays	LP in chawl (δ = 18.05%) vs harvest sale; 8 seasons	stores 0 t in 8/8 seasons (gain Rs 0); Sep–Nov rule in a chawl: Rs −1,149/t, 6/8 lost	**Not supported** (holds only if δ < 12.4%)
H3 low-loss storage pays	LP, C vs B, paired over 8 seasons	+Rs 4,546/t (median 3,840), 7/8 wins; t-test p = 0.006, Wilcoxon p = 0.008; 2023–24 +Rs 9,980/t	**Strongly supported**
H4 late rain ↑ rise (extension)	OLS on Sep+Oct rain, n = 29 seasons; LOO, placebo	≈+20% per 100 mm, p = 0.036 (HC1 0.0025), R^2 0.15; dropping 2005, 2018 or 2019 → p > 0.05; placebo p = 0.11	**Weakly supported** (suggestive)

Table 2. Hypothesis tests. The chawl plan never stores, so B equals harvest sale and C vs B equals C vs A.

H1. β = −0.442 (HAC s.e. 0.178, one-sided p = 0.007; adj. R^2 0.529): months with 10% heavier arrivals have prices about 4.4% lower, as in [10, 11]. It is an association: β weakens under other normalisations (−0.27, −0.25; p = 0.06–0.08), vanishes once last month's price enters (−0.014, p = 0.92), and last month's price predicts arrivals (p = 0.011) – supply and price are jointly determined. **Forecast.** Test MAPE 29.5%, bias −17.1%, direction right in 93.8% of

month-to-month moves: the level is off, the shape is right – and shape is what the LP needs. Flat forecasts (naive, 12-month average) earn Rs 0 because they never show a reason to store; shape-aware models earn Rs 3,980–6,182/t. **2024 plan (C).** Sell 300 t in April (capacity binds), then 21.2 t in September and 300 t each in October and November (handling cap binds); 78.8 t is lost to shrinkage. Verified marginal values: storage +Rs 4,028/t, handling +Rs 6,526 per t/month (Oct 3,051 + Nov 3,475); the loan sale *L* is slack. Planned net Rs 13,251/t realised Rs 20,813/t at actual prices, as the forecast was low.

Strategy (Rs/t vs April sale)	B chawl	C cold/IoT	C 2023–24
LP on forecast (our plan)	0 (0/8)	4,546 (7/8)	9,980
Rolling monthly re-plan	0 (0/8)	4,511 (7/8)	9,980
Calendar rule Sep–Nov (ex-ante)	−1,149 (2/8)	5,508 (7/8)	9,117
Perfect foresight (ceiling)	791 (5/8)	8,558 (8/8)	10,981

Table 3. Backtest: mean gain 2017–24 (wins of 8); last column 2023–24.

Capex of C, Rs/t capacity	Payback	Net/yr, Rs/t
8,000 (MIDH norm)	0.15 seas.	6,332
16,667 (ICAR, base)	1.49 seas.	4,921
50,000 (cold room)	6.62 seas.	−503

Table 4. Economics of C over B (10 yr, 10%).

Industry 4.0 value. Low-loss storage adds Rs 4,546/t harvested (Rs 4.55/kg), Rs 6,495 per t of capacity per season. Against Rs 9,667/t of extra capital it pays back in **1.49 seasons** and nets Rs 4,921 per t of capacity a year after capital charges: **≈Rs 45.5 lakh a year gross** for a 1,000 t FPO, ≈Rs 34.4 lakh net; it stays worthwhile up to ≈Rs 46,900/t of capital. **Sensitivity (Fig. 2b).** A chawl pays only below 12.4%/month loss (Rs 3,172/t at 6%). Cold storage stays positive at δ = 8% (Rs 374/t), at *h* = Rs 1,400 (Rs 1,069/t) and with no handling cap (Rs 4,131/t); each point of monthly loss removed is worth ≈Rs 685/t (mean slope, 0.5–8%). **Honest reading.** In cold storage the Sep–Nov rule averaged more (Rs 5,508 vs 4,546/t): the value comes mainly from the seasonal shape. The LP adapts to storage physics (the rule loses money in a chawl), won the test seasons (Rs 9,980 vs 9,117/t) and gives verified shadow prices.

Extension: weather and sensors. Wettest-third seasons rose a median 208%, driest 123% (same sign, but significant in neither subperiod). A real-time rain feed is worth **Rs 0/t**: by November C holds at most *U* = 300 t, which the cap sells then either way (B holds none) – handling capacity, not information, binds. *Loss sensing:* at the book 18% the chawl plan sells all in April, so sensing never starts; a chawl *measured* at 8% earns **Rs 2,200/t** with our forecast, but above the 12.4% break-even storing loses (14%: Rs −508/t). In cold storage our low-biased forecast (−17.1%) leaves sensing mixed; with accurate prices it never hurts: **Rs 545–746/t** if the store slips to 8–15%/month.

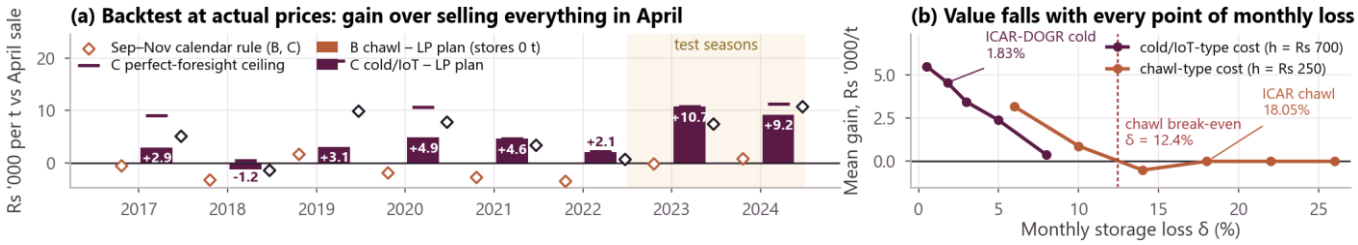


Figure 2. (a) Cold/IoT storage beats April sale in 7 of 8 seasons (2018: Rs −1,181/t); the chawl LP stores nothing. (b) Mean gain vs loss δ.

Limitations. The composite runs ≈10% below Lasalgaon, whose gains would be higher (Nasik prices: Rs 11,510 and 13,377/t in 2023–24); H1 is an association and H4 only suggestive; forecasts run low (−17%; MAPE 29.5%); loss rates extrapolate 120-day trials (ICAR's multi-module store, C's capex basis, loses ≈2.5%/month; at 3% C still gains Rs 3,444/t); grading, price impact, credit and export shocks are unmodelled; 8 seasons (2 test) make tests indicative.

8 Managerial Implications

- **FPOs.** At ICAR's 18%/month chawl loss, sell at harvest; storage pays only if curing and ventilation cut loss below ≈12% – so **measure your store's real loss rate before the season** (measured at 8%, a chawl earns Rs 2,200/t). In low-loss storage, sell the excess over capacity in April, release in Sep–Nov, re-plan monthly, and add Oct–Nov handling capacity before space (shadow prices +Rs 6,526 vs +Rs 4,028/t): logistics, not information, binds.
- **Govt buffer agencies (NAFED / NCCF).** Buying at the April trough is right, but traditional storage destroys value – the 2019 buffer loss is consistent with chawl physics. Hold buffers in cold or monitored storage and release them in Sep–Nov, when the index is 1.21–1.55× the year's mean.
- **Storage investors and policy.** Low-loss storage repays in ≈1.5 seasons at Rs 16,667/t and within one season at the MIDH Rs 8,000/t norm, but loses money after capital charges at Rs 50,000/t. Tie subsidies to loss measured by temperature, humidity and gas sensors (≈Rs 685/t per point of loss); sensors pay only with an unbiased forecast.

9 References (including the data source)

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